

Yi-Ruei Liu

Website: shigon255.github.io

Education

University of Illinois Urbana-Champaign

Master of Computer Science; GPA: 4.0/4.0

Aug. 2024 - Dec. 2025

Champaign, IL

National Yang Ming Chiao Tung University

Bachelor of Computer Science; GPA: 4.27/4.3

Sep. 2021 - Jan. 2025

Hsinchu, Taiwan

- Ranked 1st out of 167 students in the Department of Computer Science.
- Phi Tau Phi Honor Society inductee (top 1% nationwide).

Experience

Comp Photo Lab, National Yang Ming Chiao Tung University

Research Assistant (Advisor: Prof. Yu-Lun Liu)

Dec. 2025 - Present

Hsinchu, Taiwan

AIU Lab, Academia Sinica

Research Assistant (Advisor: Prof. Jun-Cheng Chen)

Jul. 2025 - Aug. 2025

Taipei, Taiwan

- Authored a research paper on synthesizing camera effects from real-world videos.
- Led the project development, including dataset construction, method implementation, and evaluation.

Comp Photo Lab, National Yang Ming Chiao Tung University

Research Assistant (Advisor: Prof. Yu-Lun Liu)

May. 2023 - Jun. 2024

Hsinchu, Taiwan

- Researched and implemented algorithms for 3D shape reconstruction from single-view images under dynamic lighting.
- Co-authored a research paper on dynamic scene reconstruction with specular objects, contributing to method development and experiment analysis.

Publications

BRDFusion: Physics Meets Generation for Urban Scene Inverse Rendering

Under Review

Yi-Ruei Liu, Jie-Ying Lee, Zheng-Hui Huang, Yu-Lun Liu, Chih-Hao Lin

- Combining physically-based rendering and generative prior to achieve high fidelity and controllable urban scene relighting.
- Achieve state-of-the-art novel-view relighting performance on synthetic benchmark, and achieve superior visual quality on real-world urban scenes.
- Contributed to main algorithm implementation, dataset construction, experiments, and analysis.

Skyfall-GS: Synthesizing Immersive 3D Urban Scenes from Satellite Imagery

ECCV 2026

Jie-Ying Lee, **Yi-Ruei Liu**, Shr-Ruei Tsai, Wei-Cheng Chang

Chung-Ho Wu, Jiewen Chan, Zhenjun Zhao, Chieh Hubert Lin, Yu-Lun Liu

- The first method to synthesize immersive, real-time navigable 3D urban scenes solely from multi-view satellite imagery.
- Enables state-of-the-art 3D urban city reconstruction with superior realism and quality.
- Contributed to core algorithm development and experiment design.

Every Camera Effect, Every Time, All at Once:

NeurIPS 2025 Workshop on SpaVLE, Oral

4D Gaussian Ray Tracing for Physics-based Camera Effect Data Generation*Yi-Ruei Liu, You-Zhe Xie, Yu-Hsiang Hsu, I-Sheng Fang, Yu-Lun Liu, Jun-Cheng Chen*

- Proposed a novel pipeline to synthesize realistic videos with physically-correct camera effects from real-world videos using dynamic 3D Gaussian Splatting and Ray Tracing.
- Achieved superior reconstruction and synthesis quality to existing baselines.
- Led the overall project development, contributed to dataset construction, method implementation, and evaluation.

SpectroMotion: Dynamic 3D Reconstruction of Specular Scenes

CVPR 2025

Cheng-De Fan, Chen-Wei Chang, Yi-Ruei Liu, Jie-Ying Lee, Jiun-Long Huang, Yu-Chee Tseng, Yu-Lun Liu

- Proposed a novel method that combines dynamic 3D Gaussian splatting with physically-based rendering for accurate reconstruction of dynamic, specular scenes.
- Achieved state-of-the-art dynamic 3D reconstruction of specular scenes.
- Contributed to core algorithm development, implementation, and experiment design.